

STEP CUE · DESIGN DECISION RECORD

StepCue recommendations accepted

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	STC-DDR-002	Version	0.1
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stepcue		
Source	docs/decisions/0002-recommendations-accepted.md @ 066cd5f		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 acceptance of all open recommendations

0002: Recommendations accepted

- **Date:** 2026-09-25
- **Status:** accepted (items O2 and O3); item O1 remains proposed

■ Context

After the TRL 3 session, `docs/REVIEW.md` and STC-DDR-001 listed three items still awaiting Amish. On 2026-09-25 Amish wrote, in chat: "i accept all your recommendations, go with them across all repos." Every open item that carried a recommendation is therefore decided as recommended. An item with no recommendation stays open. TRL 4 remains on hold by Amish's instruction, and nothing in this record moves the design past TRL 3.

■ Options considered

Table 1. Open items at the start of this decision.

#	Item	Options	Recommendation
O1	First co-design partner	Movement disorders clinic; physiotherapy practice; Parkinson's patient group	None
O2	R6 cue-start target for the phone or earbud audio route	(a) keep 0.1 s and accept that audio misses it; (b) 0.3 s for the audio route only, 0.1 s kept for the haptic default	(b)
O3	R9 insole thickness margin	(a) keep the 3.0 mm EVA base, 5.00 mm stack, zero margin; (b) 2.5 mm EVA base, 4.50 mm stack, with a 0.4 mm relief for the motor in the laminate	(b)

■ Decision

- **O2.** Decided by Amish, 2026-09-25: go with recommendation. R6 now reads: haptic cue 0.1 s or less after a detection; audio through a phone or earbuds 0.3 s or less. Latency shifts only the first beat, since the pod sends beat timing rather than a trigger per beat.
- **O3.** Decided by Amish, 2026-09-25: go with recommendation. The insole base is 2.5 mm EVA. The 2.7 mm coin motor sits on a 0.2 mm EVA floor in its pocket and stands 0.4 mm proud into a relief in the underside of the sensor laminate, leaving 0.4 mm of laminate over it.

Items still open:

- **O1.** First co-design partner. No recommendation was made. Proposed, awaiting Amish.

No budget, pitch or problem change was recommended, so `budget_usd` stays at \$200 and the pitch and problem lines are unchanged. No item needs a change in another repo. No item called for TRL 4 work.

■ Consequences

Table 2. What changed in the repo.

Item	File	Before	After
O2	docs/03-requirements.md (STC-REQ-001 v0.4)	R6: 0.1 s for any cue; audio route not met (0.18 to 0.28 s)	R6: 0.1 s haptic, 0.3 s audio route; met (41 ms haptic, 0.18 to 0.28 s audio)
O2	docs/04-calcs/sizing.py , 01-sizing.md (STC-CAL-001 v0.2)	R6 status "Not met (audio route)"	Two targets checked; R6 met with 20 ms margin at worst on assumed audio latencies
O3	cad/src/model.py , cad/step/ , cad/stl/	foam_t 3.0 mm; motor top flush with the EVA	foam_t 2.5 mm; motor on a 0.2 mm floor; 0.4 mm relief cut in the laminate
O3	docs/03-requirements.md , STC-CAL-001	R9 at risk, 5.00 mm, zero margin	R9 met, 4.50 mm (4.3 to 4.7 mm with EVA tolerance)
O3	cad/drawings/STC-DWG-001	Rev P1, stack 5.0	Rev P2, stack 4.5, motor relief noted
O3	bom/bom.csv	Line 6: 3 mm EVA	Line 6: 2.5 mm EVA; line 3 notes the motor relief; total unchanged at \$83.44
O2, O3	docs/02-concept.md (STC-PRC-001 v0.4)	O2 and O3 listed as proposed	Recorded as decided; key numbers updated
O2, O3	docs/decisions/0001-trl2-review-decisions.md (STC-DDR-001 v0.2)	O2 and O3 "Proposed, awaiting Amish"	Marked decided, pointing to this record
O3	media/	3.0 mm base	Regenerated from the updated model

Requirement status at TRL 3 (STC-CAL-001 v0.2) moves from 9 met, 1 not met, 4 at risk and 1 not verifiable to 11 met, 0 not met, 3 at risk (R3, R4, R5) and 1 not verifiable (R12).

The design stops at TRL 3. Measuring the audio-route latency, the insole thickness and the felt motor intensity through the thinner stack is TRL 4 work and is on hold.